



MURANG'A UNIVERSITY OF TECHNOLOGY

SCHOOL OF COMPUTING AND INFORMATION TECHNOLOGY

DEPARTMENT OF INFORMATION TECHNOLOGY

UNIVERSITY ORDINARY EXAMINATION

2022/2023 ACADEMIC YEAR

THIRD YEAR SECOND SEMESTER EXAMINATION FOR BSE,BMCS,BIT&BCT

SCS 302: ARTIFICIAL INTELLIGENCE

DURATION: 2 HOURS

INSTRUCTIONS TO CANDIDATES:

1. Answer Question one and Any other two questions.
2. Mobile phones are not allowed in the examination room.
3. You are not allowed to write on this examination question paper.

SECTION A

QUESTION ONE: COMPULSORY

- a) Define the following term
 - i. Artificial Intelligence (1 mark)
 - ii. Fuzzy Logic (1 mark)
 - iii. Agent (1 mark)
 - iv. Backward chains (1 mark)
 - v. Environment (1 mark)
- b) The intelligence is intangible. Give elements that compose intelligence according to A.I (5 marks)
- c) Discuss the two goals of artificial intelligence (4 marks)
- d) Discuss tuning test in A.I (5 marks)
- e) Explain the difference between Human Machine Intelligence (6 marks)
- f) The expert systems are computer application developed to solve complex problems in a particular domain. Explain five capabilities of expert systems (5 marks)

SECTION B: ANSWER ANY TWO QUESTIONS.

QUESTION TWO:

- a) Learning enhances the awareness of the discipline of the study. Discuss the various categories of learning in artificial intelligence. (10 marks)
- b) Discuss at least five real life applications of Artificial Intelligence (5 marks)

QUESTION THREE:

- a) Explain the main four components of fuzzy logic systems architecture (8 marks)
- b) Explain the difference between Human & machine Intelligence (4 marks)
- c) Explain three discipline applicable in A.I (3 marks)

QUESTION FOUR:

- a) Natural language processing refers to AI method for communicating with intelligent systems. Discuss the five steps of NLP (10 marks)
- b) Explain Five Capabilities of expert systems (5 marks)